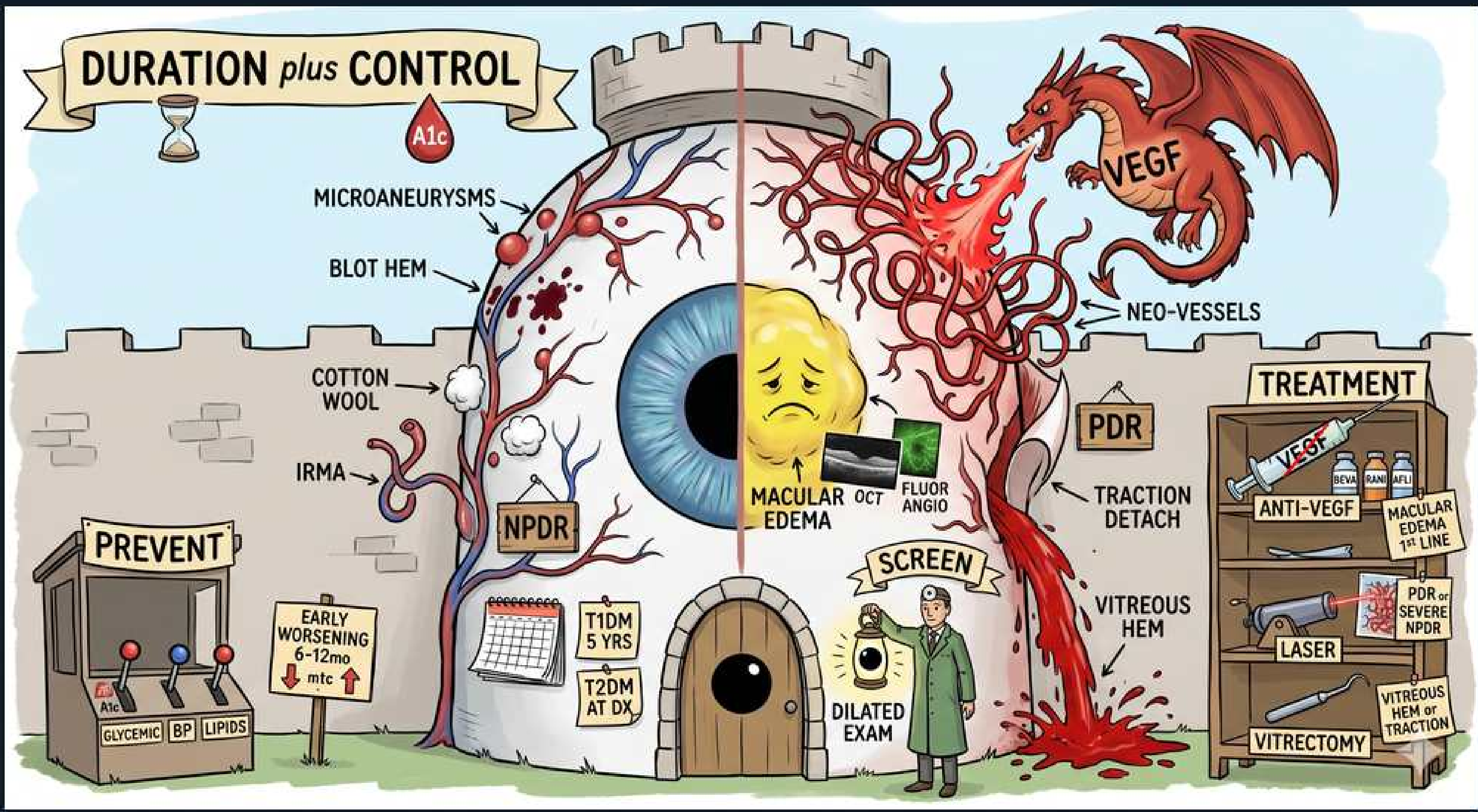
DM — Scene 1 — Diabetic Complications: The Eye Tower



1. Duration + A1c banner

WHAT YOU SEE

Hourglass and A1c blood drop on the banner header

WHAT IT ENCODES

The two strongest determinants of diabetic retinopathy are disease DURATION and chronic glycemic control (A1c); hypertension is the major additional modifiable accelerator. Nearly all T1DM and >60% of T2DM patients have some retinopathy by ~20 years. Each ~1% drop in A1c meaningfully reduces microvascular progression (DCCT/UKPDS). Mnemonic: 'time + sugar build the retinopathy.'

BOARD TRAP

WRONG to think tight SHORT-TERM control reverses ESTABLISHED retinopathy — control prevents/slows progression but does not undo existing structural damage, and rapid correction can transiently worsen it. Discriminator: the drivers are cumulative duration and chronic A1c, not a single acute glucose value.

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2. Microaneurysms

WHAT YOU SEE

Tiny red dots labeled MICROANEURYSMS on the tower wall

WHAT IT ENCODES

Microaneurysms — small outpouchings of weakened retinal capillary walls — are the EARLIEST and hallmark finding of non-proliferative diabetic retinopathy (NPDR), appearing as tiny discrete red dots on fundus exam. They reflect pericyte loss and capillary wall fragility from chronic hyperglycemia. Mnemonic: 'micro-aneurysm = the first dot of NPDR.'

BOARD TRAP

Neovascularization is the WRONG answer for the FIRST/earliest sign of retinopathy — neovessels define PROLIFERATIVE disease (PDR), a late stage. Discriminator: the earliest detectable lesion is the microaneurysm of NPDR, not new vessel growth.

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3. Blot hemorrhage

WHAT YOU SEE

Dark red blot labeled BLOT HEM

WHAT IT ENCODES

Dot-and-blot intraretinal hemorrhages are deeper hemorrhages (in the inner nuclear/outer plexiform layers) seen in NPDR; flame hemorrhages, by contrast, are superficial nerve-fiber-layer bleeds. Their increasing number marks worsening NPDR severity. Mnemonic: 'dot-blot = deep; flame = superficial.'

BOARD TRAP

WRONG to interpret dot-and-blot hemorrhages as proliferative disease — they are NPDR features. Discriminator: it is NEOVASCULARIZATION (not the quantity of hemorrhages) that defines PDR; numerous hemorrhages indicate severe NPDR at high risk of progressing.

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4. Cotton wool spots

WHAT YOU SEE

Fluffy white spot labeled COTTON WOOL

WHAT IT ENCODES

Cotton wool spots are fluffy white retinal lesions representing nerve-fiber-layer MICROINFARCTS from capillary occlusion (axoplasmic flow stasis); their appearance signals worsening (pre-proliferative) NPDR and ischemia. They also occur in hypertension and HIV retinopathy. Mnemonic: 'cotton wool = soft ischemic infarct.'

BOARD TRAP

WRONG to equate cotton wool spots with hard exudates — cotton wool spots are SOFT, fluffy ischemic microinfarcts, whereas HARD exudates are sharp, yellow LIPID/protein deposits from leaky vessels. Discriminator: fluffy white + indistinct = cotton wool (ischemia); waxy yellow + well-defined = hard exudate (lipid leakage).

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5. IRMA

WHAT YOU SEE

Branching vessel labeled IRMA

WHAT IT ENCODES

Intraretinal microvascular abnormalities (IRMA) are dilated, tortuous shunt vessels within the retina that mark SEVERE NPDR and confer high risk of progression to PDR. They represent attempts at collateral flow around occluded capillaries (not true neovascularization). The '4-2-1 rule' defines severe NPDR (hemorrhages in 4 quadrants, venous beading in 2, or IRMA in 1). Mnemonic: 'IRMA = inside the retina, almost proliferative.'

BOARD TRAP

WRONG to classify IRMA as neovascularization/PDR — IRMA are INTRAretinal shunts of severe NPDR, while neovessels grow ON the retinal/disc SURFACE in PDR. Discriminator: IRMA stay within the retina and do not leak on angiography the way fragile new vessels do; their presence means severe NPDR (treat aggressively, screen closely), not yet PDR.

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6. NPDR placard

WHAT YOU SEE

Sign reading NPDR below the eye

WHAT IT ENCODES

NPDR (non-proliferative diabetic retinopathy) is the stage defined by microaneurysms, dot-blot hemorrhages, cotton wool spots, hard exudates, venous beading, and IRMA — but NO neovascularization. Mild/moderate NPDR is managed with risk-factor control and closer screening; severe NPDR may warrant early panretinal photocoagulation or anti-VEGF given high PDR risk. Mnemonic: 'NPDR = leaky but not yet sprouting.'

BOARD TRAP

WRONG to think NPDR never threatens vision and only needs routine follow-up — diabetic macular edema (the leading cause of vision loss) can occur during NPDR, and severe NPDR has high progression risk. Discriminator: presence of center-involving macular edema or severe NPDR features changes management from observation to active treatment.

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7. Macular edema / OCT

WHAT YOU SEE

Swollen yellow macula labeled MACULAR EDEMA with OCT and fluorescein angio icons

WHAT IT ENCODES

Diabetic macular edema (DME) — fluid accumulation/retinal thickening at the macula from breakdown of the blood-retinal barrier — is the LEADING cause of vision loss in diabetes. Optical coherence tomography (OCT) is the diagnostic test of choice to detect and quantify thickening; fluorescein angiography maps leakage. First-line treatment for center-involving DME is intravitreal ANTI-VEGF. Mnemonic: 'DME blurs central vision — OCT to see it, anti-VEGF to treat it.'

BOARD TRAP

WRONG to assume DME only occurs in proliferative disease — DME can develop at ANY stage (NPDR or PDR). Discriminator: vision loss in a diabetic with only NPDR is most often DME, diagnosed by OCT, and anti-VEGF (not laser) is first-line for center-involving edema.

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8. VEGF dragon / Neo-vessels

WHAT YOU SEE

Red dragon breathing fire labeled VEGF with NEO-VESSELS

WHAT IT ENCODES

Progressive retinal ISCHEMIA upregulates vascular endothelial growth factor (VEGF), which drives growth of fragile, leaky NEOVESSELS on the retina, optic disc (NVD), and iris — the defining feature of proliferative diabetic retinopathy (PDR). These vessels bleed easily and contract, causing hemorrhage and traction. Mnemonic: 'ischemia breathes VEGF, VEGF grows the dragon's new vessels.'

BOARD TRAP

WRONG to think new vessels are protective collaterals to be preserved — neovessels are fragile, leak, bleed (vitreous hemorrhage), and cause tractional detachment. Discriminator: anti-VEGF and panretinal photocoagulation aim to ABLATE/suppress neovascularization, not encourage it; untreated VEGF-driven neovascularization can also cause neovascular glaucoma.

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9. PDR placard

WHAT YOU SEE

Sign reading PDR

WHAT IT ENCODES

Proliferative diabetic retinopathy (PDR) is defined by NEOVASCULARIZATION (of disc, retina, or iris) and carries high risk of severe vision loss from vitreous hemorrhage, tractional retinal detachment, and neovascular glaucoma. Treatment is panretinal photocoagulation (PRP) and/or anti-VEGF; vitrectomy for non-clearing hemorrhage or traction. Mnemonic: 'PDR = Proliferating, Dangerous, needs Rapid treatment.'

BOARD TRAP

WRONG to manage PDR with observation/risk-factor control alone — it requires active ophthalmologic treatment (PRP and/or anti-VEGF). Discriminator: the appearance of any neovascularization upgrades NPDR to PDR and mandates urgent referral, not routine annual follow-up.

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10. Traction detachment / vitreous hem

WHAT YOU SEE

Labels TRACTION DETACH and VITREOUS HEM with blood pooling

WHAT IT ENCODES

PDR's fragile neovessels rupture into the vitreous (vitreous hemorrhage → sudden floaters/vision loss) and form fibrovascular membranes that contract, producing TRACTIONAL retinal detachment — both causes of acute, severe vision loss. Management may require anti-VEGF, PRP, or vitrectomy. Mnemonic: 'new vessels bleed and pull — hemorrhage then traction.'

BOARD TRAP

WRONG to assume sudden painless vision loss in a PDR patient is a stroke/optic event without considering vitreous hemorrhage or tractional detachment — these are the classic PDR complications. Discriminator: sudden floaters/curtain with known PDR points to vitreous hemorrhage or tractional detachment; urgent dilated exam/ultrasound differentiates them.

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11. Screen / dilated exam

WHAT YOU SEE

Doctor with magnifier labeled SCREEN, DILATED EXAM

WHAT IT ENCODES

Screening for diabetic retinopathy is a DILATED fundus exam (or validated retinal photography): T1DM beginning 5 YEARS after diagnosis; T2DM AT the time of diagnosis (subclinical disease is often already present); then at least ANNUALLY (less often if exams are repeatedly normal, more often if disease or pregnancy). Pregnant patients with pre-existing diabetes need an exam in the first trimester and close follow-up. Mnemonic: 'T1 in 5, T2 at diagnosis, then yearly.'

BOARD TRAP

Screening T2DM only 5 years after diagnosis (like T1DM) is the WRONG answer — T2DM is screened AT diagnosis because hyperglycemia (and retinopathy) often predates clinical detection. Discriminator: the 5-year delay applies to T1DM; T2DM gets a dilated exam at diagnosis.

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12. Early worsening warning

WHAT YOU SEE

Sign EARLY WORSENING 6–12mo with up arrow

WHAT IT ENCODES

Rapid intensification of glycemic control can cause TRANSIENT early worsening of retinopathy at ~6–12 months (from hemodynamic/IGF-1 shifts), especially with very high baseline A1c or advanced retinopathy. Long-term, tight control is clearly beneficial — so do not avoid control, but examine the retina before/after rapid correction and monitor closely. Mnemonic: 'tighten control, watch the eyes for 6–12 months.'

BOARD TRAP

WRONG to AVOID lowering A1c because of early-worsening risk — long-term tight control reduces retinopathy progression; the worsening is transient. Discriminator: the correct action is a baseline dilated exam plus closer monitoring during rapid correction, not withholding glycemic improvement.

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13. Prevent booth (control)

WHAT YOU SEE

Booth labeled PREVENT with GLYCEMIC, BP, LIPIDS levers

WHAT IT ENCODES

Prevention and slowing of retinopathy rests on the modifiable risk factors: glycemic control (lower A1c), blood pressure control, and lipid management; smoking cessation also helps. Hypertension control is particularly impactful for retinopathy/maculopathy progression (UKPDS). Mnemonic: 'sugar, pressure, lipids — control all three.'

BOARD TRAP

WRONG to think glucose control alone is sufficient prevention — BP control independently slows retinopathy progression, and untreated hypertension worsens both retinopathy and macular edema. Discriminator: comprehensive risk-factor control (glucose + BP + lipids) outperforms focusing on A1c alone.

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14. Treatment shelf (anti-VEGF / laser / vitrectomy)

WHAT YOU SEE

Shelf labeled TREATMENT: ANTI-VEGF, LASER, VITRECTOMY

WHAT IT ENCODES

Treatment tiers: intravitreal ANTI-VEGF (ranibizumab, aflibercept, bevacizumab) is first-line for center-involving diabetic macular edema and is also used for PDR; panretinal photocoagulation (LASER) is mainstay for PDR and severe NPDR; focal/grid laser is an option for non-center DME; VITRECTOMY is reserved for non-clearing vitreous hemorrhage or tractional retinal detachment. Mnemonic: 'anti-VEGF for the macula, laser for the periphery, vitrectomy for the mess.'

BOARD TRAP

WRONG to choose focal/grid LASER as first-line for CENTER-involving macular edema — anti-VEGF is first-line there (better visual outcomes). Discriminator: anti-VEGF leads for center-involving DME, while panretinal LASER remains a primary option for proliferative disease/severe NPDR.
